21 $p^3 \times p^x = p^9$

(a) Find the value of x.

$x =$

(1)

$(7^2)^y = 7^{10}$

(b) Find the value of y.

$y =$

(1)

$100^a \times 1000^b$ can be written in the form 10^w

(c) Show that $w = 2a + 3b$

(2)

(Total for Question 21 is 4 marks)